Vectors

Vector Basics

Definition: A **vector** has both magnitude (size) and direction. A **scalar** has only magnitude. Vectors are written as $\mathbf{a}$, $\vec{AB}$, or in column form $\begin{pmatrix} x \\ y \end{pmatrix}$.

In 2D, a vector can be written as $\mathbf{a} = x\mathbf{i} + y\mathbf{j} = \begin{pmatrix} x \\ y \end{pmatrix}$.

In 3D, a vector can be written as $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

Magnitude

$$|\mathbf{a}| = \sqrt{x^2 + y^2} \quad (2\text{D}) \qquad |\mathbf{a}| = \sqrt{x^2 + y^2 + z^2} \quad (3\text{D})$$

Unit Vectors

Unit Vector: A vector of magnitude 1 in the direction of $\mathbf{a}$:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

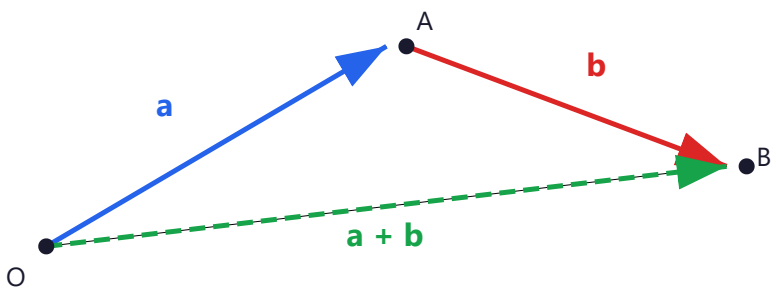
Vector Operations

Addition and Subtraction

Vectors are added component-wise:

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$$

Triangle Law of Addition



Scalar Multiplication

Multiplying vector $\mathbf{a}$ by scalar k scales its magnitude by $|k|$. If $k < 0$, the direction reverses.

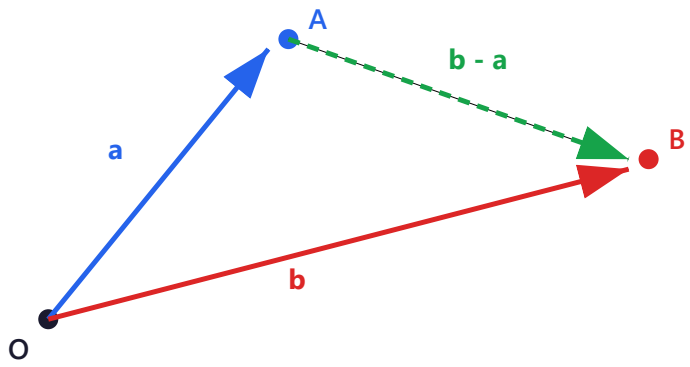
Position Vectors

Position Vector: The position vector of a point A is $\vec{OA}$, the vector from the origin O to A. If $A = (a_1, a_2, a_3)$, then $\vec{OA} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$.

The vector from A to B is:

$$\vec{AB} = \vec{OB} - \vec{OA} = \mathbf{b} - \mathbf{a}$$

Position Vectors: AB = b - a



Worked Example: Distance Between Points

Find the distance between $A(3, 1, -2)$ and $B(1, -1, 4)$.

Solution:

$$\vec{AB} = \begin{pmatrix} 1-3 \\ -1-1 \\ 4-(-2) \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \\ 6 \end{pmatrix}$$

$$|\vec{AB}| = \sqrt{(-2)^2 + (-2)^2 + 6^2} = \sqrt{4+4+36} = \sqrt{44} = 2\sqrt{11}$$

Scalar Product (Dot Product)

Scalar Product: For vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$$

where θ is the angle between them. In component form:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Key properties:

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ (commutative)
- If $\mathbf{a} \cdot \mathbf{b} = 0$, the vectors are **perpendicular**
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

Angle Between Two Vectors

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Worked Example: Angle Between Vectors

Find the angle between $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$.

Solution:

$$\mathbf{a} \cdot \mathbf{b} = (2)(1) + (1)(-1) + (-1)(3) = 2 - 1 - 3 = -2$$

$$|\mathbf{a}| = \sqrt{4 + 1 + 1} = \sqrt{6}, \quad |\mathbf{b}| = \sqrt{1 + 1 + 9} = \sqrt{11}$$

$$\cos\theta = \frac{-2}{\sqrt{6}\sqrt{11}} = \frac{-2}{\sqrt{66}} \implies \theta = \cos^{-1}\left(\frac{-2}{\sqrt{66}}\right) = 104.3^\circ$$

Exam Tip: If the dot product is negative, the angle is obtuse (greater than 90 degrees). Always give angles to 1 decimal place unless told otherwise. Remember that the angle between two vectors is always between 0 and 180 degrees.

Geometric Problems with Vectors

Midpoint

The midpoint M of AB has position vector:

$$\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$$

Dividing a Line Segment

The point dividing AB in the ratio $m : n$ has position vector:

$$\vec{OP} = \frac{n\mathbf{a} + m\mathbf{b}}{m + n}$$

Worked Example: Proving Perpendicularity (with coordinates)

Points $A(1, 2, 3)$, $B(3, 0, 1)$, and $C(4, 2, 0)$. Show that angle $ABC = 90^\circ$.

Solution: Find vectors from B:

$$\vec{BA} = \begin{pmatrix} 1-3 \\ 2-0 \\ 3-1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}, \quad \vec{BC} = \begin{pmatrix} 4-3 \\ 2-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\vec{BA} \cdot \vec{BC} = (-2)(1) + (2)(2) + (2)(-1) = -2 + 4 - 2 = 0$$

Since $\vec{BA} \cdot \vec{BC} = 0$, the vectors are perpendicular, so angle $ABC = 90^\circ$.

Worked Example: Proving Perpendicularity

Show that $\mathbf{p} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{q} = 2\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ are perpendicular.

Solution:

$$\mathbf{p} \cdot \mathbf{q} = (3)(2) + (1)(-4) + (-2)(1) = 6 - 4 - 2 = 0$$

Since $\mathbf{p} \cdot \mathbf{q} = 0$, the vectors are perpendicular.

Exam Tip: To prove vectors are parallel, show that one is a scalar multiple of the other. To prove perpendicularity, show the dot product is zero. These are the two most common vector proof questions at A-Level.

Vector Equation of a Line

Vector Equation of a Line: A straight line through point A with position vector $\mathbf{a}$, in direction $\mathbf{d}$, has equation:

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$$

where λ is a scalar parameter. Every point on the line corresponds to a unique value of λ.

To write the line through two points A and B:

- Direction vector: $\mathbf{d} = \vec{AB} = \mathbf{b} - \mathbf{a}$
- Line equation: $\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$

Worked Example: Equation of a Line

Find the vector equation of the line through $A(2, 1, -3)$ and $B(4, 0, 1)$.

Solution:

$$\vec{AB} = \begin{pmatrix} 4-2 \\ 0-1 \\ 1-(-3) \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

Checking if a Point Lies on a Line

A point P lies on $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$ if there exists a consistent value of λ satisfying all three component equations simultaneously.

Worked Example: Point on a Line

Does $C(8, -2, 13)$ lie on $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$?

Solution: From the x-component: $2 + 2\lambda = 8 \implies \lambda = 3$.

Check y: $1 + (-1)(3) = -2$. ✓

Check z: $-3 + 4(3) = 9 \neq 13$. ✗

The λ values are inconsistent, so C does not lie on the line.

Collinearity

Three points A, B, C are **collinear** if $\vec{AC} = k\,\vec{AB}$ for some scalar k — i.e. the vectors are parallel and share a common point.

Worked Example: Proving Collinearity

Show that $A(1, 2, 3)$, $B(3, 5, 7)$ and $C(5, 8, 11)$ are collinear.

Solution:

$$\vec{AB} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad \vec{AC} = \begin{pmatrix} 4 \\ 6 \\ 8 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 2\,\vec{AB}$$

Since $\vec{AC} = 2\,\vec{AB}$, the vectors are parallel and share point A, so A, B, C are collinear.

Intersection and Skew Lines

In 3D, two lines can either **intersect**, be **parallel**, or be **skew** (non-parallel and non-intersecting).

To test two lines $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$ and $\mathbf{r} = \mathbf{b} + \mu\mathbf{e}$:

1. Set $\mathbf{a} + \lambda\mathbf{d} = \mathbf{b} + \mu\mathbf{e}$ and write out three component equations
2. Solve two of the equations for λ and μ
3. Substitute into the third equation — if satisfied, the lines intersect; otherwise they are skew

Exam Tip: Always check whether direction vectors are scalar multiples of each other first — if they are, the lines are parallel (or the same line). Only if they are not parallel do you need to test for intersection vs. skew.

Angle Between Two Lines

The acute angle θ between lines with direction vectors $\mathbf{d}_1$ and $\mathbf{d}_2$:

$$\cos\theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$$

The modulus signs ensure the acute angle is always returned.

Worked Example: Angle Between Lines

Find the acute angle between lines with directions $\mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.

Solution:

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 3 - 2 - 2 = -1$$

$$|\mathbf{d}_1| = \sqrt{6}, \quad |\mathbf{d}_2| = \sqrt{14}$$

$$\cos\theta = \frac{|-1|}{\sqrt{6}\sqrt{14}} = \frac{1}{\sqrt{84}} \implies \theta = \cos^{-1}\left(\frac{1}{\sqrt{84}}\right) \approx 83.7^\circ$$